

Post-quantum transition for device bounded anonymous credentials: a EUDI wallet application

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Abstract

In 2026, member states of the European Union are expected to have running their own EUDI wallet applications instances compliant with the ARF architecture guidelines. The goal of the ARF architecture is to provide privacy preserving anonymous credentials to users, but the current implementation fails. In the meantime, in 2024 the European Union has published a roadmap for the post quantum digital transition of members digital infrastructures.

No current implementation of anonymous credentials is post quantum secure at the moment, with cryptologist and mathematician in top publications claiming that is not possible to build a post quantum anonymous credential scheme with the actual technologies, giving hardware constraints and advanced requirements. So we asked ourselves: is it possible to build a post quantum device bounded anonymous credential scheme with the actual technologies, giving hardware constraints and technical difficulties? The answer is **yes**, it is possible.

Moreover, in this thesis we present a post quantum anonymous credentials model that can be deployed with the actual technologies and legacy infrastructures, with NIST standardized post quantum signature schemes. We went further and implemented the full model and run it into android devices in order to benchmark the performance and the results are surprising. We also present some optimizations that could be implemented in order to further improve the performance of the scheme, and we show how to implement them with the actual technologies.

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1 Introduction

The european union on 11 April 2024 has published a roadmap for the post quantum digital transition of members digital infrastructures.

1.1 Overview

2 State of the art

2.1 Anonymous Credentials

2.2 Post quantum anonymous credentials

3 Implementation